Enrollment No._____

# GANPAT UNIVERSITY

B.Sc.IT(Data Science) Sem-II Remedial Nov - Dec 2024

U72A1IP2: INTRODUCTION TO PROGRAMMING-II

Time: 3 Hours Total: 60 Marks

**Instructions:**

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

| **Section – I** | | | | |
|---|---|---|---|---|
| **Q-1** | **Answer the following. (Any 5)** | **Marks 30** | **CO** | **Bloom's Taxonomy** |
| 1 | Discuss compile time initialization of one dimensional array with example. | 06 | CO1 | BTL6 |
| 2 | Explain String Handling functions with example. | 06 | C01 | BTL5 |
| 3 | Discuss Scope, Visibility and Lifetime of variables with example. | 06 | CO2 | BTL5 |
| 4 | What is Pointer? Discuss pointer and array with example. | 06 | CO4 | BTL3 |
| 5 | Write a short note on Union. Differentiate structure V/S union. | 06 | CO3 | BTL4 |
| 6 | How to use command line argument in file I/O operation ? Discuss it. | 06 | CO4 | BTL6 |
| 7 | Discuss malloc(),calloc() and realloc() function. | 06 | CO5 | BTL5 |
| **Section – II** | | | | |
| **Q-2** | **Answer the following (Any Two)** | **Marks 12** | **CO** | **Bloom's Taxonomy** |
| 1 | Write a short note on Two-Dimensional Array. | 06 | CO1 | BTL3 |
| 2 | Explain getchar( ), gets( ), putchar( ), and puts( ) functions. | 06 | CO1 | BTL5 |
| 3 | Explain Nesting of functions with elaborating UDF Elements. | 06 | CO2 | BTL5 |
| | | | | |
| **Q.3** | **Answer the following: (Any Two)** | **12** | | |
| 1 | Discuss array of structure with example. | 06 | CO3 | BTL6 |
| 2 | What is the difference between call by value and call by reference? Discuss it with example. | 06 | CO4 | BTL4 |
| 3 | Discuss fprintf() and fscanf() function with example. | 06 | CO4 | BTL5 |

| | | | | |
|---|---|---|---|---|
| | | | | |
| **Q.4** | **Answer the following: (Any One)** | **06** | | |
| 1 | Write a short note on Preprocessor. | 06 | CO5 | BTL3 |
| 2 | Discuss Common programming Errors. | 06 | CO5 | BTL3 |

**** End of the Paper ****

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# GANPAT UNIVERSITY

B.Sc. IT (Data Science) Sem-II Remedial Nov - Dec 2024

U72A2DA2 DATA ANALYSIS-II

Time: 3 Hours Total: 60 Marks

## Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

| **Section – I** | | | | |
|---|---|---|---|---|
| **Q-1** | **Answer the following. (Any 5)** | **Marks 30** | **CO** | **Bloom's Taxonomy** |
| 1 | Evaluate the Statistical functions with an example. | 06 | CO1 | BTL5 |
| 2 | Evaluate the Database functions with an example. | 06 | CO1 | BTL5 |
| 3 | What is a Chart? List out various charts. Discuss any two charts in detail. | 06 | CO2 | BTL2/BTL6 |
| 4 | Demonstrate the Goal Seek function. | 06 | CO3 | BTL3 |
| 5 | What is PivotTable? Write down the steps for creating PivotTables from external Data. | 06 | CO4 | BTL2/BTL3 |
| 6 | Write down the steps to protect worksheets and workbooks. | 06 | CO5 | BTL3 |
| 7 | What is Hyperlink? How to create Hyperlinks? Discuss with an example. | 06 | CO5 | BTL2/BTL6 |
| **Section – II** | | | | |
| **Q-2** | **Answer the following (Any Two)** | **Marks 12** | **CO** | **Bloom's Taxonomy** |
| 1 | Evaluate the Financial functions with an example. | 06 | CO1 | BTL5 |
| 2 | Classify the SmartArt diagram with an example | 06 | CO2 | BTL4 |
| 3 | What is a Sparkline? Create with an example. | 06 | CO2 | BTL2/BTL6 |
| | | | | |
| **Q.3** | **Answer the following: (Any Two)** | **12** | | |
| 1 | Analyze Data validation tools in detail. | 06 | CO4 | BTL4 |
| 2 | Develop the Scenario Manager with an example | 06 | CO3 | BTL6 |
| 3 | Write a short note on Macro. | 06 | CO3 | BTL3 |
| | | | | |
| **Q.4** | **Answer the following: (Any One)** | **06** | | |
| 1 | Write down the steps to adding Headers and Footers to Printed Pages. | 06 | CO5 | BTL3 |
| 2 | Discuss different setting options while printing a workbook or worksheet. | 06 | CO5 | BTL3 |

**** End of the Paper ****

Enrollment No._____

# GANPAT UNIVERSITY

B.Sc. IT (DATA SCIENCE) Sem-II Remedial Nov - Dec 2024

U72A3DMS DATABASE MANAGEMENT SYSTEM

Time: 3 Hours Total: 60 Marks

## Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

| Section – I | | | | |
|---|---|---|---|---|
| **Q-1** | **Answer the following. (Any 5)** | **Marks 30** | **CO** | **Bloom's Taxonomy** |
| 1 | What are Dr. E. F. Codd's rules for RDBMS and why are they important? | 06 | CO1 | BTL1 |
| 2 | Highlight the distinctions between DBMS and RDBMS, emphasizing their architectural variances, storage mechanisms, and operational disparities. | 06 | CO1 | BTL2 |
| 3 | How does the E/R Model facilitate database design, and what role do E/R Diagrams play in this process? | 06 | CO2 | BTL3 |
| 4 | Explore the capabilities of the Select clause in DQL commands. Discuss how it is used to retrieve specific data from one or more tables in a database. | 06 | CO3 | BTL5 |
| 5 | Analyze the different types of joins supported in Oracle SQL.Explain the syntax and usage of each type, along with scenarios where they are most appropriate. | 06 | CO3 | BTL4 |
| 6 | Explore the usage of the Rollup and Cube operators. How do these operators enhance data summarization and analysis? Provide scenarios where each operator would be beneficial. | 06 | CO5 | BTL3 |
| 7 | Discuss the significance of indexes in Oracle databases. How do they improve query performance, and what factors should be considered when deciding which columns to index? | 06 | CO5 | BTL2 |
| **Section – II** | | | | |
| **Q-2** | **Answer the following (Any Two)** | **Marks 12** | **CO** | **Bloom's Taxonomy** |
| 1 | Describe the duties of a DBA. | 06 | CO1 | BTL5 |

**[P.T.O]**

| | | | | |
|---|---|---|---|---|
| 2 | Define normalization and explain the significance of trivial and non-trivial dependencies in database design. Outline the steps to achieve First, Second, and Third Normal Forms, emphasizing dependency preservation. | 06 | CO2 | BTL1 |
| 3 | Discuss Boyce/Codd Normal Form, Multi-Valued Dependencies, Fourth Normal Form and Fifth Normal Form, with practical examples. | 06 | CO2 | BTL2 |
| | | | | |
| **Q.3** | **Answer the following: (Any Two)** | **12** | | |
| 1 | Discuss the Logical Operators and Range Searching Operators. Provide examples demonstrating their usage in database queries. | 06 | CO3 | BTL2 |
| 2 | Explore SQL predicates. Discuss how they are used to filter and retrieve data based on specific criteria in SQL queries. | 06 | CO3 | BTL3 |
| 3 | Discuss the role of the Group By clause in conjunction with aggregate functions. How does grouping data affect the results produced by aggregate functions, and what considerations should be made when using both together? | 06 | CO3 | BTL2 |
| | | | | |
| **Q.4** | **Answer the following: (Any One)** | **06** | | |
| 1 | Explain the functionality of the Dump function in Oracle databases. How can the Dump function be used to analyze and debug data storage formats? | 06 | CO5 | BTL5 |
| 2 | Explain the concept of spooling in Oracle databases. How can spooling be used to capture query results and save them to a file for later analysis? | 06 | CO5 | BTL6 |

**** End of the Paper ****

Enrollment No._____

# GANPAT UNIVERSITY

B.Sc (Data Science) Sem-II Remedial Nov - Dec 2024

U72A4COA COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours Total: 60 Marks

**Instructions:**

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

| **Section – I** | | | | |
|---|---|---|---|---|
| **Q-1** | **Answer the following. (Any 5)** | **Marks 30** | **CO** | **Bloom's Taxonomy** |
| 1 | What are decimal, binary, octal, and hexadecimal number systems? Explain each briefly. | 06 | CO1 | BTL1, BTL2 |
| 2 | How do you detect errors in digital data using error detection codes? | 06 | CO1 | BTL4 |
| 3 | Contrast CISC (Complex Instruction Set Computers) with RICS. What are the advantages and disadvantages of each? | 06 | CO2 | BTL5 |
| 4 | Discuss the significance of the Excess-3 code, binary to gray code and gray to binary code. | 06 | CO2 | BTL6 |
| 5 | Describe the architecture and operation of an array processor in detail. | 06 | CO3 | BTL4 |
| 6 | Describe 8085 microprocessor pin diagram. | 06 | CO4 | BTL5 |
| 7 | Elaborate Assembly Programming Instruction in brief. | 06 | CO5 | BTL3, BTL4 |
| **Section – II** | | | | |
| **Q-2** | **Answer the following (Any Two)** | **Marks 12** | **CO** | **Bloom's Taxonomy** |
| 1 | What is the significance of using 1's and 2's complement methods in binary addition and subtraction? | 06 | CO1 | BTL2, BTL3 |
| 2 | Write down difference between ASCII CODE & EBCDIC CODE. | 06 | CO2 | BTL5, BTL6 |
| 3 | Elaborate RICS Characteristics, advantages and disadvantages. | 06 | CO2 | BTL4 |
| | | | | |
| **Q.3** | **Answer the following: (Any Two)** | **12** | | |
| 1 | Compare and contrast the various addressing techniques in brief. | 06 | CO3 | BTL5 |

| | | | | |
|---|---|---|---|---|
| 2 | Describe the operation of a processor level with multifunction. | 06 | CO3 | BTL4 |
| 3 | Explain architecture of 8085 block diagram. | 06 | CO4 | BTL5 |
| | | | | |
| **Q.4** | **Answer the following: (Any One)** | **06** | | |
| 1 | Describe Arithmetic Instruction and Logical instruction. | 06 | CO5 | BTL4 |
| 2 | Explain the purpose of the IN and OUT instructions in assembly programming language. Provide examples of their usage. | 06 | CO5 | BTL5 |

**** End of the Paper ****

Enrollment No._____

# GANPAT UNIVERSITY

B.Sc .IT(DS) Sem-II Remedial Nov - Dec 2024

U72A5BS-BASIC STATISTICS

Time: 3 Hours Total: 60 Marks

## Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern:BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

### Section – I

| Q-1 | Answer the following. (Any 5) | Marks 30 | CO | Bloom's Taxonomy |
|---|---|---|---|---|
| 1 | The following data gives the daily wages of workers in a manufacturing company. Find the Median wage.<br>Daily wage (in 1000 Rs) : 6 8 10 12 15 18 19<br>No. of workers : 20 14 7 16 12 2 3 | 06 | CO1 | BTL5 |
| 2 | The table below shows the distribution of marks obtained by 50 students in mathematics. Find the Median marks. | 06 | CO1 | BTL5 |
| 3 | The following data Represents the profit (in lakhs) of 50-bisuness men in a large city. Find Quartile deviation of the data.<br>Profit : 20-30 30-40 40-50 50-60 60-70<br>No. of Business men : 8 12 20 7 3 | 06 | CO1 | BTL4,5 |
| 4 | The number of motor cycle accident cases reported in a hospital is as follows. Find the standard deviation for the data.<br>Age : 0-10 10-20 20-30 30-40 40-50 50-60<br>No. of Accidents : 15 49 37 20 6 1 | 06 | CO1 | BTL2,5 |
| 5 | Find Karl Pearson's co-efficient of correlation for the following data. | 06 | CO2 | BTL2,5 |
| 6 | Find Regression co-efficient of Y on X for following bivariate frequency distribution. | 06 | CO3 | BTL4,6 |

| Marks | 10-14 | 15-19 | 20-24 | 25-29 |
|---|---|---|---|---|
| No. of student | 4 | 6 | 10 | 5 |

| 30-34 | 35-39 | 40-44 | 45-49 |
|---|---|---|---|
| 7 | 3 | 9 | 6 |

| X | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
|---|---|---|---|---|---|---|---|
| Y | 6 | 9 | 12 | 15 | 18 | 21 | 24 |

| X→ / Y↓ | 0-10 | 10-20 | 20-30 |
|---|---|---|---|
| 20-30 | 2 | 3 | 0 |
| 30-40 | 1 | 2 | 1 |
| 40-50 | 1 | 2 | 3 |

7 A fair die is rolled. Find the probability of the following events: 06 CO4 BTL5

(a) A prime number will appear.
(b) A Number greater than or equal to 3 will appear.
(c) A number more than 6 will appear.
(d) A number less than 6 will appear.

## Section-II

| Q-2 | Answer the following (Any Two) | Marks 12 | CO | Bloom's Taxonomy |
|---|---|---|---|---|
| 1 | For his research on the living standard, a researcher conducted a survey on 100 persons. The distribution of the ages of the person is given below. Find the modal Age.<br>Age : 0-10 10-20 20-30 30-40 40-50 50-60<br>No. of People : 10 12 15 17 12 13 | 06 | CO1 | BTL2,5 |
| 2 | Find Harmonic mean for the following data.<br>Marks : 10-20 20-30 30-40 40-50 50-60<br>No. of Students : 6 14 22 7 1 | 06 | CO1 | BTL2,5 |
| 3 | Find the Co-efficient of variance for the following data. | 06 | CO1 | BTL5 |

| Class | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 |
|---|---|---|---|---|---|
| f | 4 | 5 | 3 | 2 | 1 |

| Q.3 | Answer the following: (Any Two) | 12 | CO | BTL |
|---|---|---|---|---|
| 1 | Find Karl Pearson's co-efficient of correlation form the following Bivariate frequency distribution. | 06 | CO2 | BTL6 |

| Marks in hindi | Marks in Mathematics | | | |
|---|---|---|---|---|
| | 0-10 | 10-20 | 20-30 | 30-40 |
| 0-10 | 4 | 2 | 0 | 1 |
| 10-20 | 1 | 3 | 2 | 0 |
| 20-30 | 1 | 1 | 3 | 4 |
| 30-40 | 4 | 0 | 0 | 1 |

| | | | | |
|---|---|---|---|---|
| 2 | Find co-efficient of correlation and probable error for the following data. | 06 | CO2 | BTL2,5 |

| X | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| Y | 5 | 6 | 3 | 2 | 4 |

| | | | | |
|---|---|---|---|---|
| 3 | Find Regression line of Y on X for the following data. Also estimate the value of Y for X=10. | 06 | CO3 | BTL2,6 |

| X | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| Y | 10 | 20 | 25 | 30 |

| Q.4 | Answer the following: (Any One) | 06 | | |
|---|---|---|---|---|
| 1 | A bag contains 4 white and 3 black balls. Two draws of 2 balls are successively made. What is the probability of getting 2 white balls at first draw and 2 black balls at second draw when balls drawn at first draw were replaced ? | 06 | CO4 | BTL5 |
| 2 | Form a pack of 52 cards two cards are drawn, the first is replaced before the second is drawn. Find the probability that the first one is diamond and the second one is a king. | 06 | CO4 | BTL5 |

**** End of the Paper ****